Akshaya S

251901012

Ex. No.: 1

CAPTURE FLAGS-ENCRYPTION CRYPTO 101

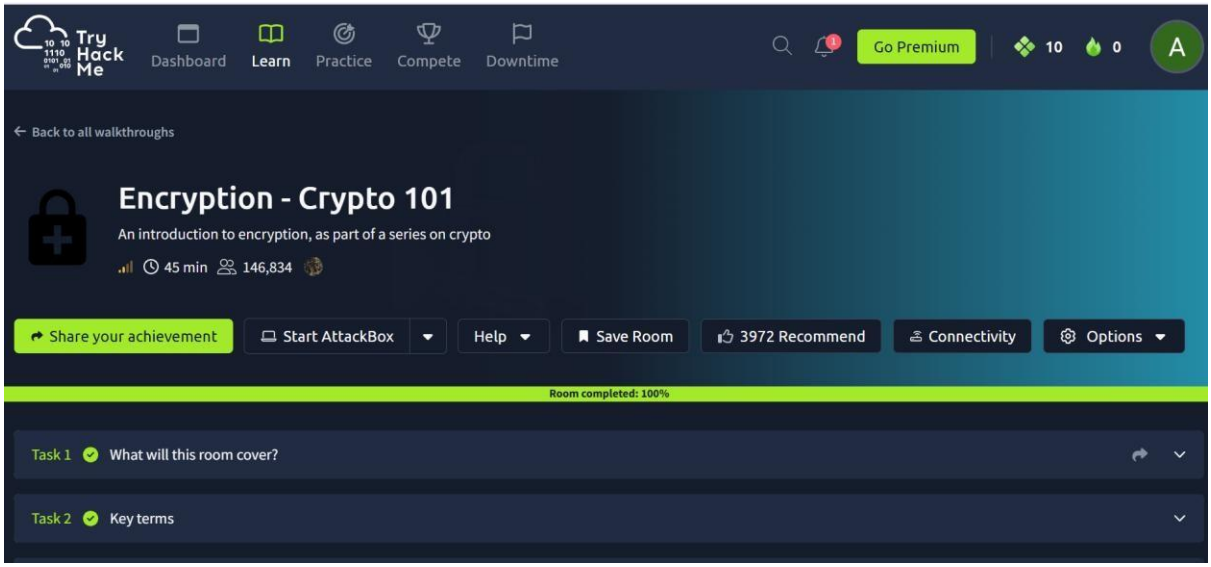
Aim:

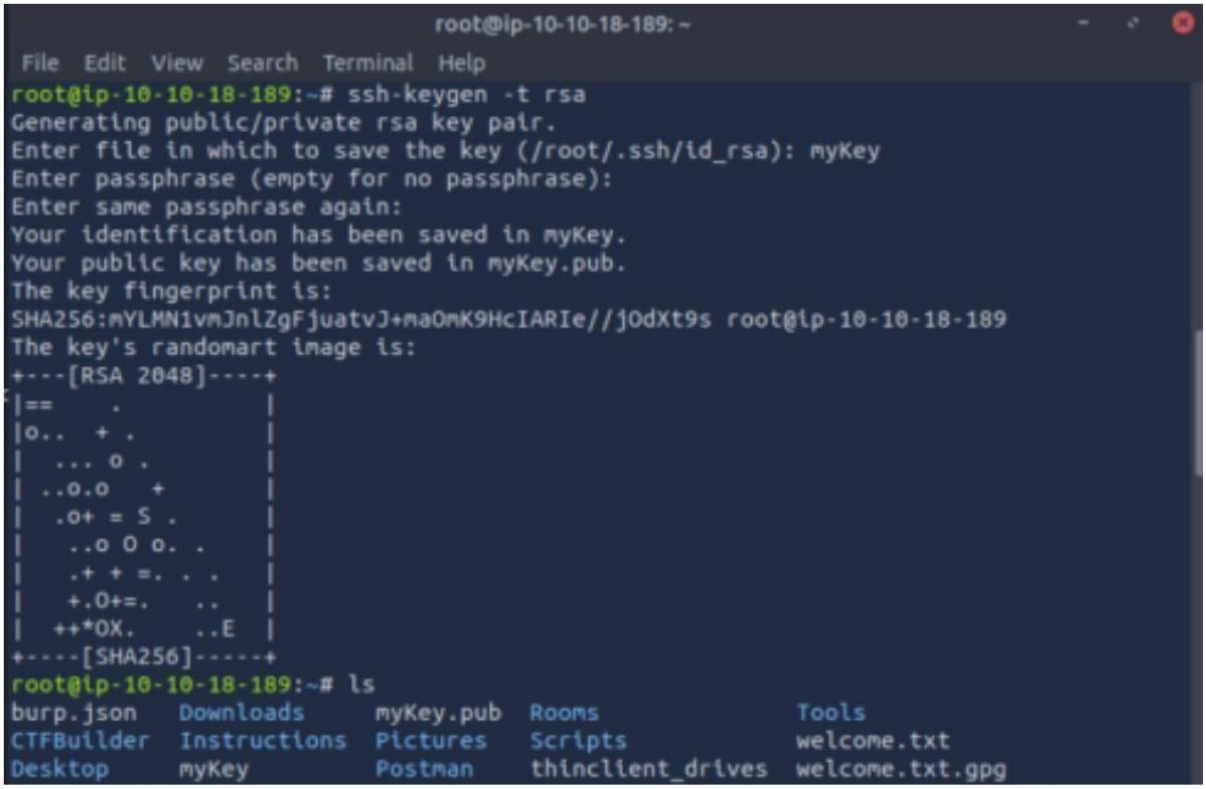
To capture the various flags in Encryption Crypto 101 in TryHackMe platform.

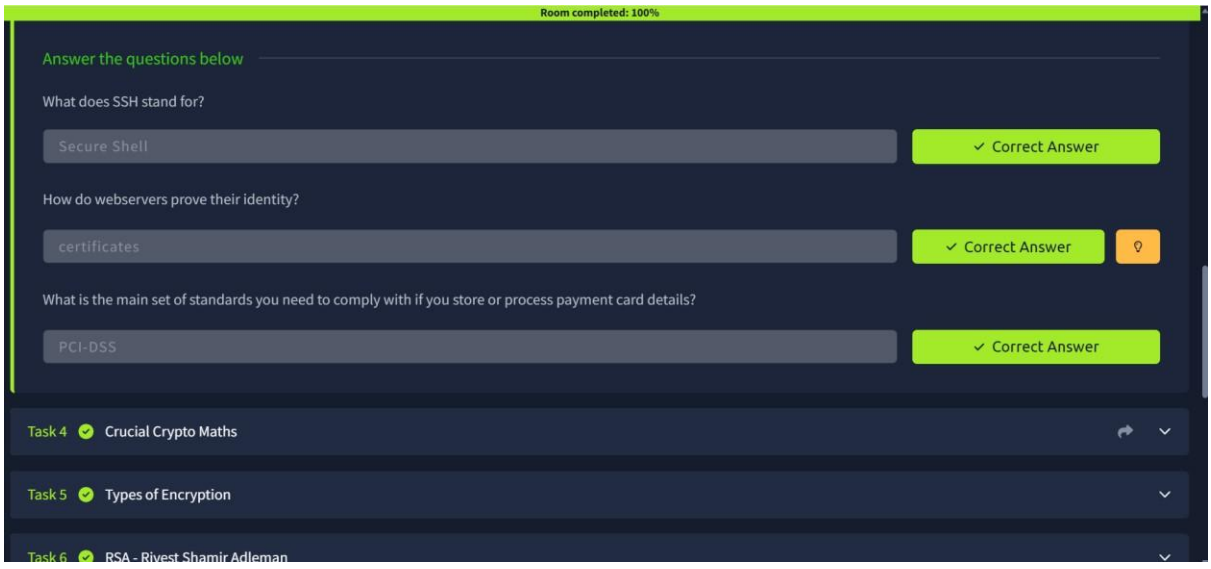
Algorithm:

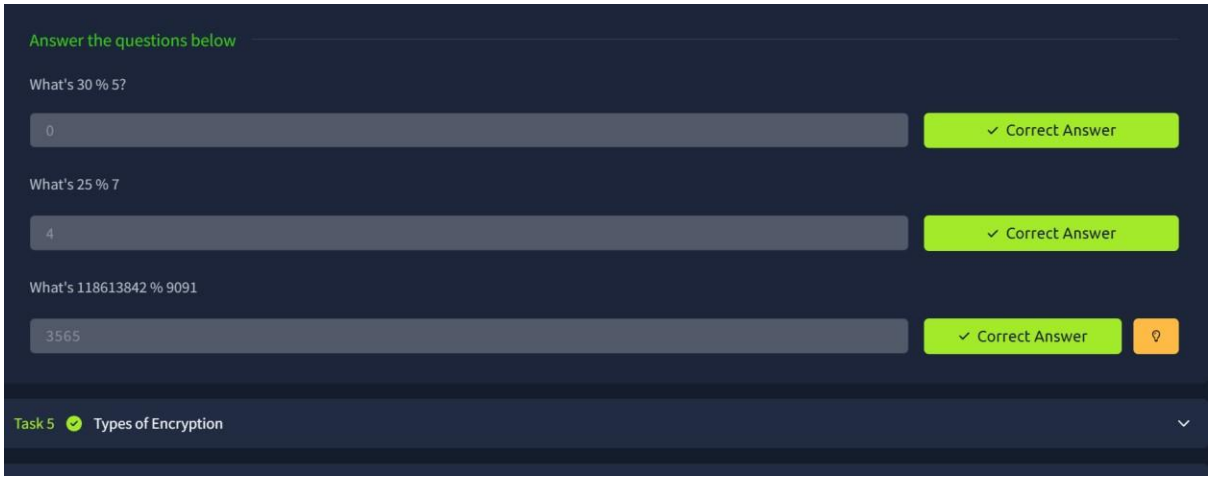
1. Access the Passive reconnaissance lab in TryHackMe platform using the link below https://tryhackme.com/r/room/encryptioncrypto101
2. Click Start AttackBox to run the instance of Kali Linux distribution.
3. Solve the crypto math used in RSA.
4. Find out who issued the HTTPS Certificate to tryhackme.com
5. Perform SSH Authentication by generating public and private key pair using sshkeygen 6. Perform decryption of the gpg encrypted file and find out the secret word.

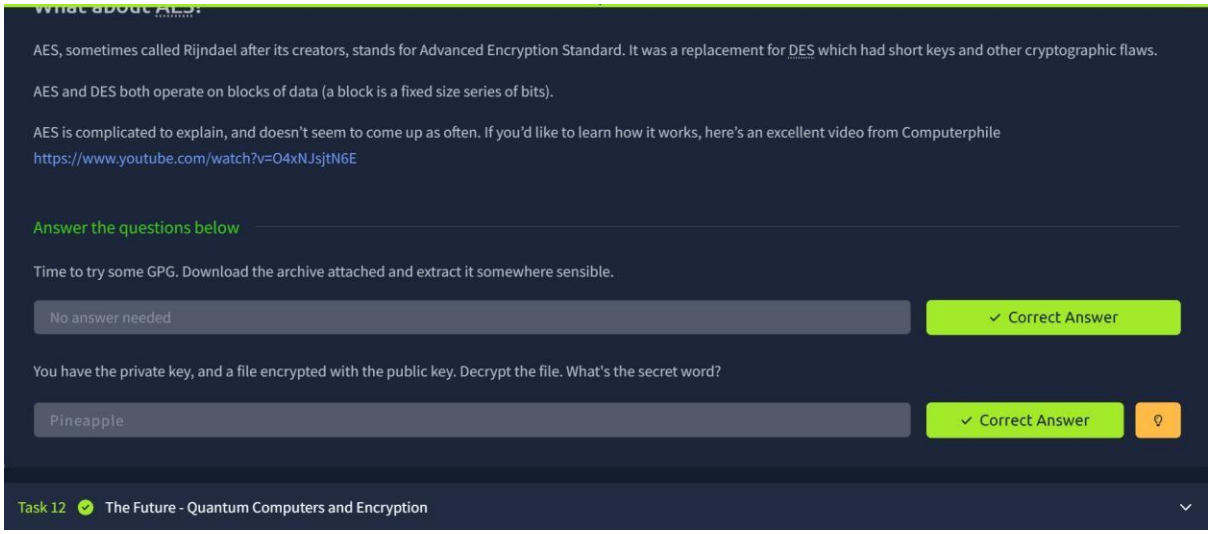
Output:




root@ip-10-10-18-189: ~
File Edit View Search Terminal Help
root@ip-10-10-18-189:~# ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa): myKey
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in myKey.
Your public key has been saved in myKey.pub.
The key fingerprint is:
SHA256:mYLMN1vmJnlZgFjuatvJ+maOmK9HcIARIe//jOdXt9s root@ip-10-10-18-189
The key's randomart image is:
+---[RSA 2048]----+
|==   . |
|o.. + . |
| ... o . |
| ..o.o   + |
| .o+ = S . |
| ..o O o. . |
| .+ + =. . . |
| +.O+=.   .. |
| ++*OX.   ..E |
+----[SHA256]-----+
root@ip-10-10-18-189:~# ls
burp.json Downloads myKey.pub Rooms Tools
CTFBuilder Instructions Pictures Scripts welcome.txt
Desktop myKey Postman thinclient_drives welcome.txt.gpg


Room completed: 100%
Answer the questions below
What does SSH stand for?
Secure Shell
Correct Answer
How do webservers prove their identity?
certificates
Correct Answer
What is the main set of standards you need to comply with if you store or process payment card details?
PCI-DSS
Correct Answer
Task 4 Crucial Crypto Maths
Task 5 Types of Encryption
Task 6 RSA - Rivest Shamir Adleman





root@ip-10-10-18-189:~# gpg --import tryhackme.key gpg:

/root/.gnupg/trustdb.gpg: trustdb created

gpg: key FFA4B5252BAEB2E6: public key "TryHackMe (Example Key)" imported gpg: key FFA4B5252BAEB2E6: secret key imported

gpg: Total number processed: 1 gpg:

imported: 1 gpg: secret keys read: 1 gpg:

secret keys imported: 1 root@ip-10-10-18-

189:~# gpg message.gpg gpg: WARNING: no

command supplied. Trying to guess what you

mean ... gpg: encrypted with 1024-bit RSA

key, ID 2A0A5FDC5081B1C5, created 2020-

06-

30 "TryHackMe (Example Key)" gpg: WARNING: no command supplied. Trying to guess what you mean ... gpg: encrypted with 1024-bit RSA key, ID 2A0A5FDC5081B1C5, created 2020-0630 "TryHackMe (Example Key)"

Result: Thus, the various flags have been captured in Encryption Crypto 101 in TryHackMe platform.